Drop BD perpendicular to the tangent — this is the "subtense of the angle of contact," the precise measure of how far the curve has risen above AD. Now the clever auxiliary: draw $BG \perp AB$ and $AG \perp AD$, meeting at G. Because both $\angle ABG$ and $\angle AGD$ are right, the points A, B, G lie on a circle with AG as its diameter. A chord dropped from that circle gives the exact relation $AB^2 = AG \times BD$ — trapping the tiny departure BD inside a clean, computable ratio.